Steps for Wake Word Change and Configuration Operations

The default wake word for this open-source project is "Hello Xiaozhi", which was applied for by the project leader/group administrator through the official Espressif channels.

If you want to change the wake word, there are currently two options:

🍰 How to **change the wake word**

1. ESP-IDF Environment Support

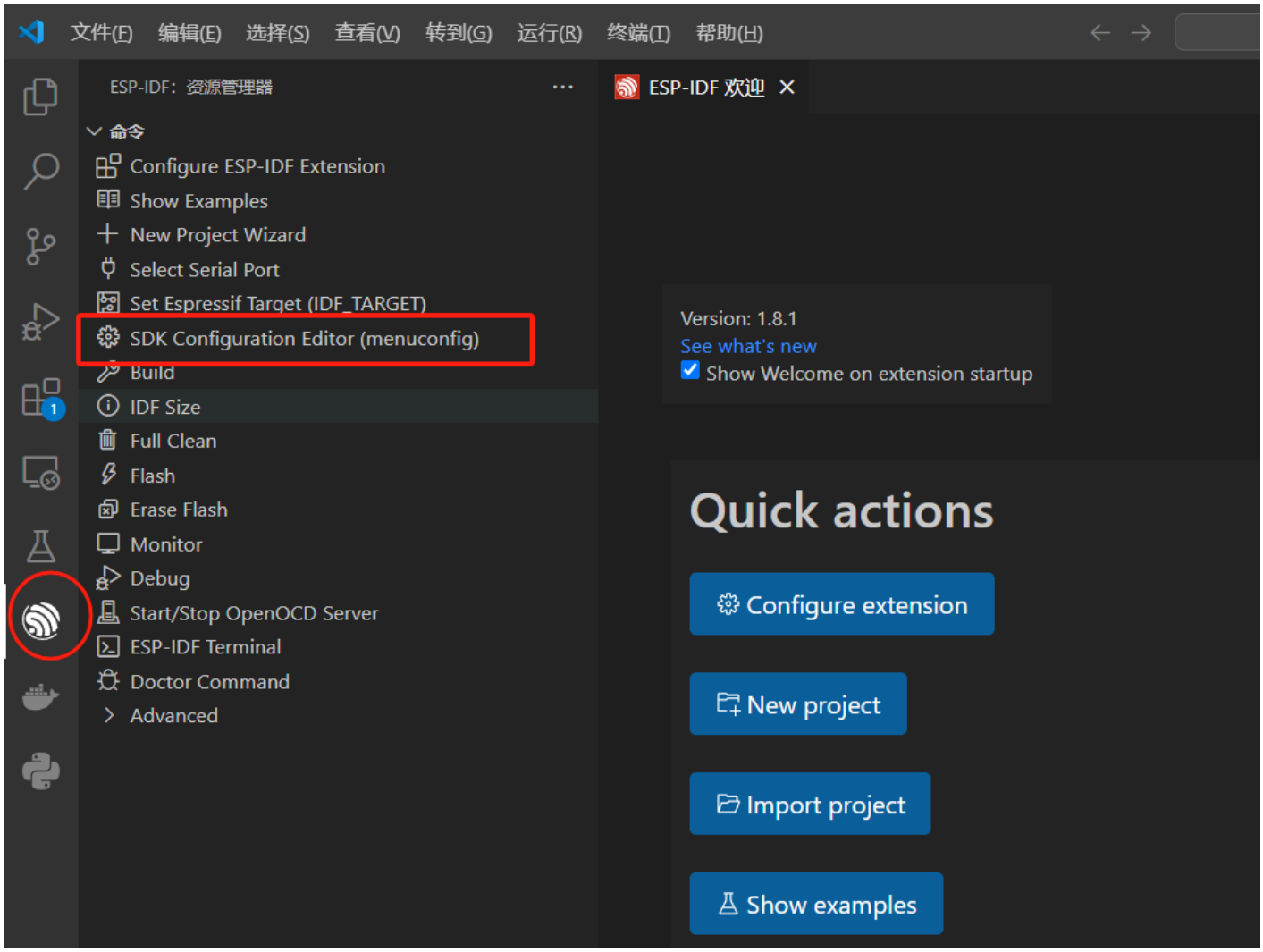
For AI voice projects, Espressif officially provides support for modifying some freely available wake words (about 26), but this needs to be configured under the installed and configured ESP IDF Development Environment and in an environment where the xiaozhi-esp32 source code can be compiled normally.

Note: ESP-IDF Environment Support is a prerequisite for modifying the wake word. If the IDF Development Environment is not properly configured, please first visit the Espressif official website to learn about the relevant configuration instructions, which are omitted here.

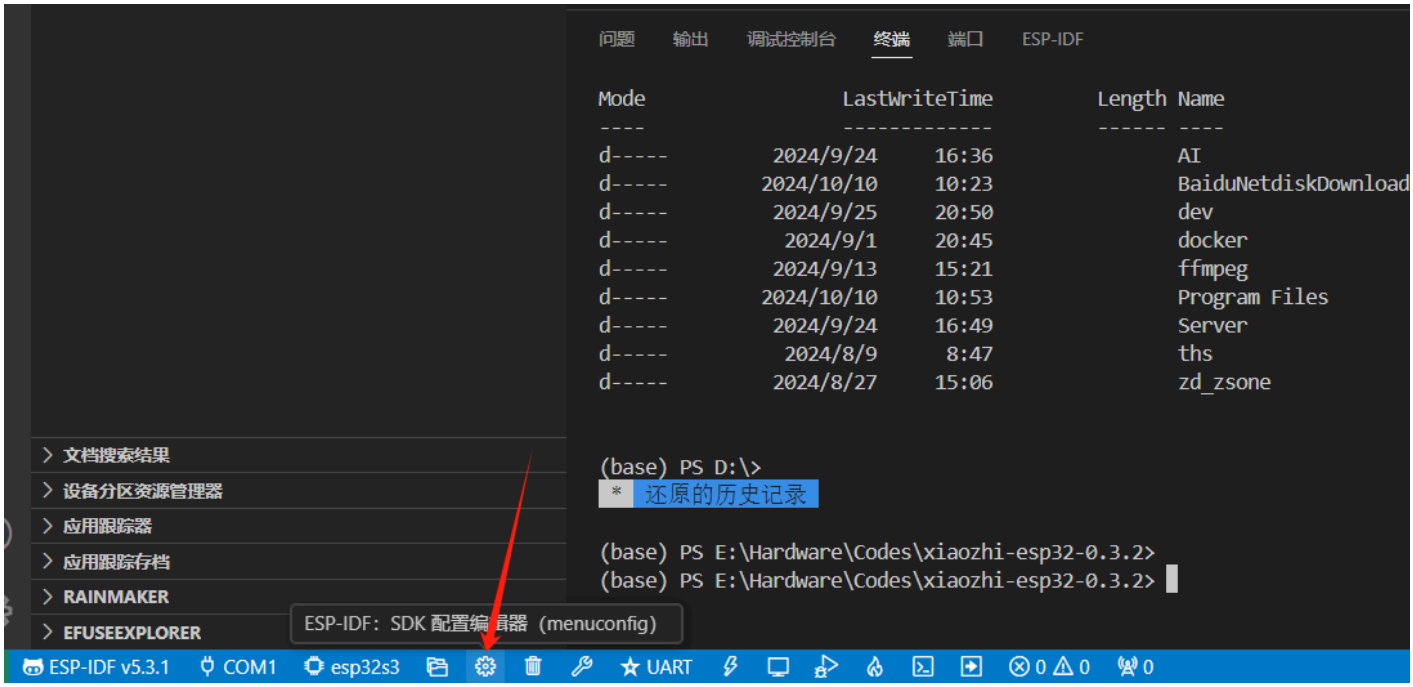
2. Method 1: Modify via the VSCode view

2.1 **SDK Configuration Entry**

Open VSCode, as shown in the figure below, find the installed ESP-IDF: Explorer icon on the left navigation bar, click to view commands, select SDK Configuration Editor (menuconfig), and clicking will take you to the SDK configuration page (the first time you enter, you need to wait for the initial configuration to load).

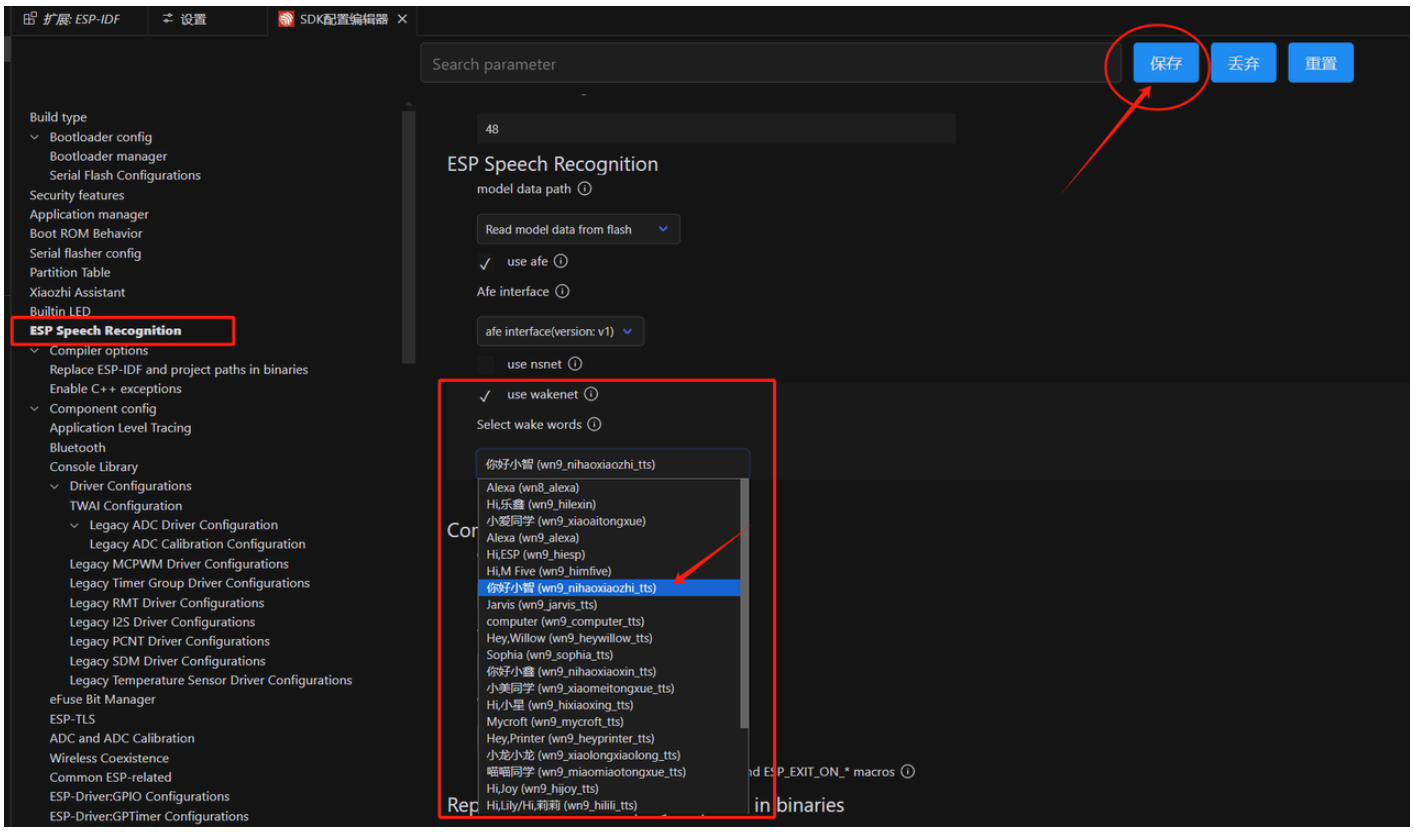


Or click the configuration icon in the bottom status bar to directly enter, as shown in the following figure:



2.2 SDK Configuration Operations

On the left side of the SDK Configuration page, click ESP Speech Recognition. In the configuration content on the right, find use wakenet (already checked in this project). Under Select wake words, select the corresponding wake word from the dropdown box to modify, make a selection according to your preference, as shown in the figure below, and finally click**Save**Settings.



> Note: The above are wake word selection options for the ESP32-S3 as the target chip. If it is ESP32 or other chips, there will only be a few wake words. Please first switch the target chip to esp32s3 (USB-JTAG) in the VSCode software status bar.

2.3 Recompile and Burn

First, close all configuration options, ensure that the ESP32S3 development board is properly connected to the computer, and it is generally recommended **to connect to the UART interface** , which corresponds to the right interface when the two TypeC interfaces on the front of the development board face downward.

- **Device COM Port Number** corresponds to the COM port selected in the status bar **(plug icon in the status bar)**
- **Device target is esp32s3 (status bar chip icon)**
- **The burning method is UART (star icon in the status bar)**
- **Build the flashing operation,** click the status bar below **ESP-IDF: Build, Flash, and Monitor icon (flame icon** , as shown by the arrow in the figure below), to rebuild, compile,

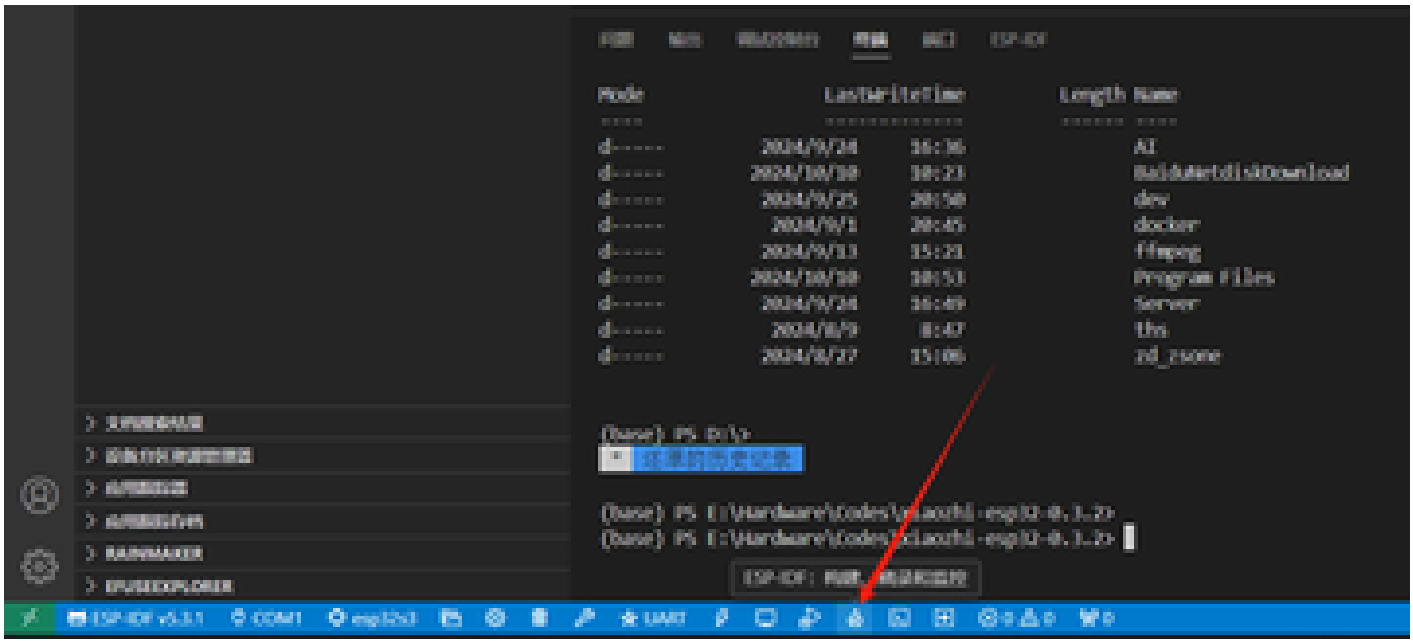
and flash (the first build may take some time, please be patient; if an error occurs, you can check it yourself according to the prompt or consult in the QQ group).

Wait for the successful completion of the burning process. After seeing the device automatically restart in the monitoring window, use the new wake word to wake up the device, thus completing the wake word replacement operation.

***Note:**

If the project has been compiled previously, to ensure updates, you may choose to use idf.py fullclean to clear the cached components.

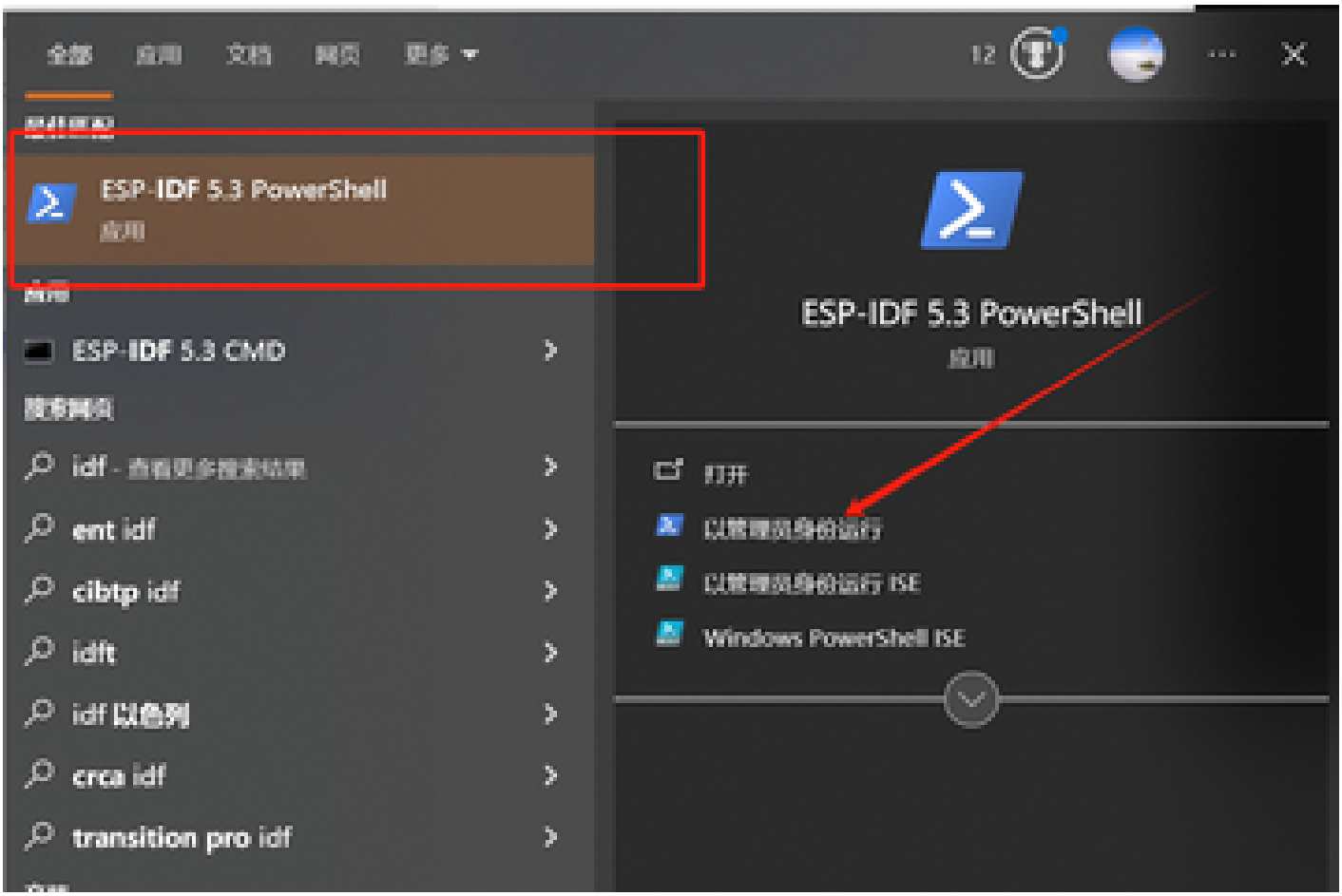
You can also choose to first build the project and then flash the firmware to the ESP32S3 development board. Please choose the operation at your own discretion.



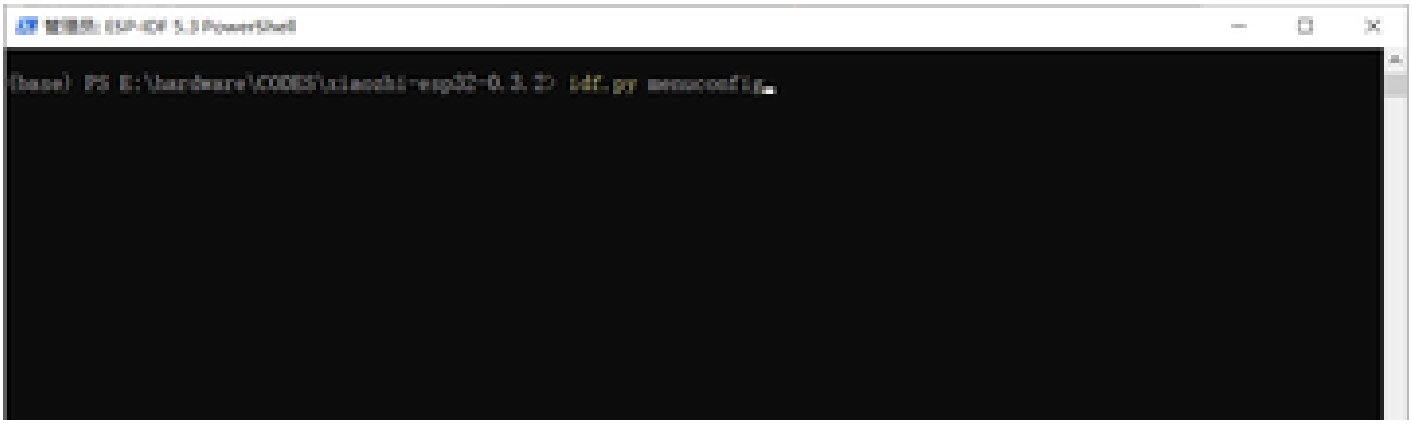
3. Method 2: Modify via the idf.py menuconfig menu

3.1 Entry command idf.py menuconfig

Please ensure that the ESP-IDF Development Environment has been installed and configured on your Windows computer. Open ESP-IDF 5.3 PowerShell from the computer's Start menu (it is recommended to run it as an administrator; ESP-IDF 5.3 CMD can also be used), as shown in the following figure:

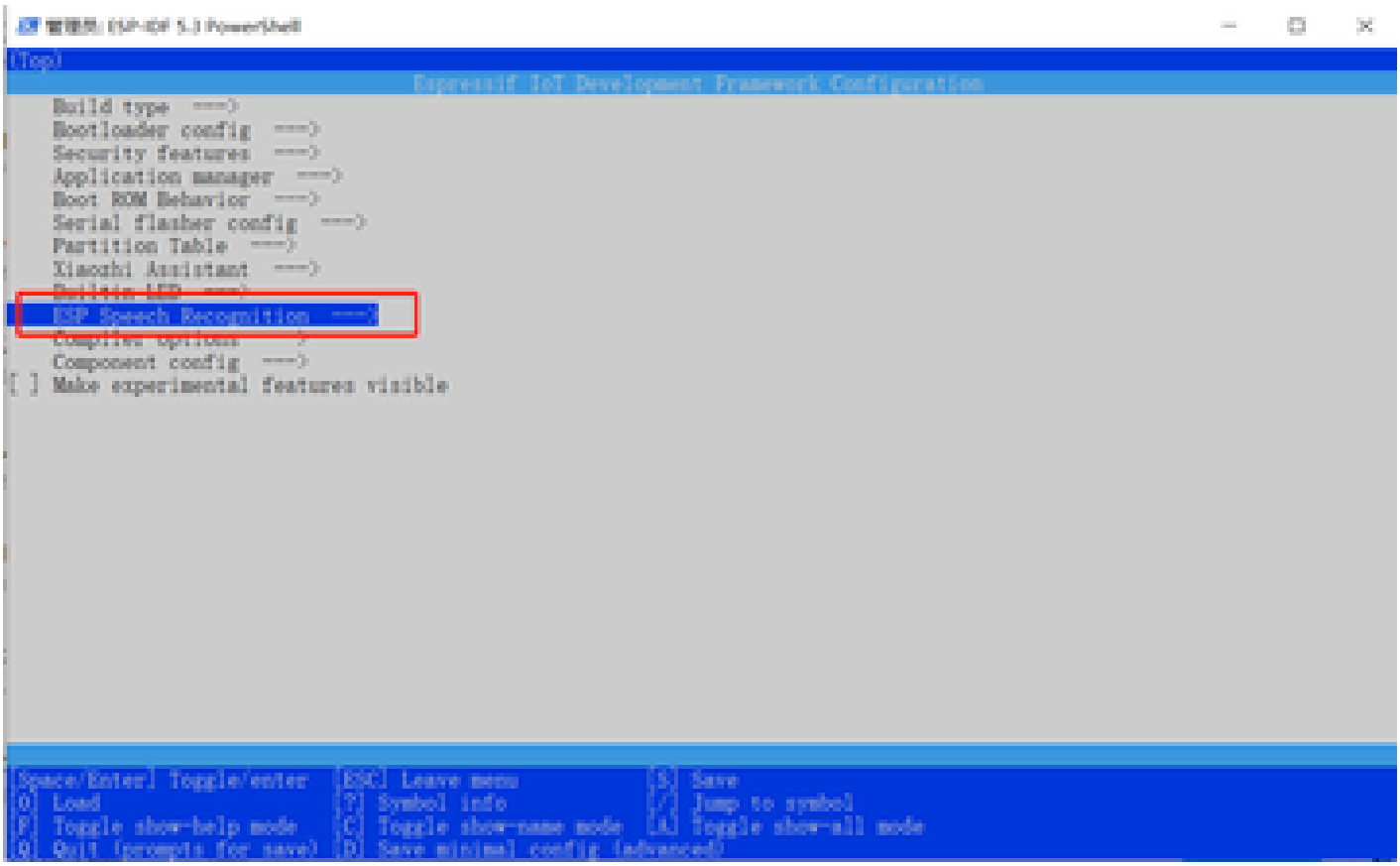


Enter idf.py menuconfig in the ESP-IDF 5.3 PowerShell command line



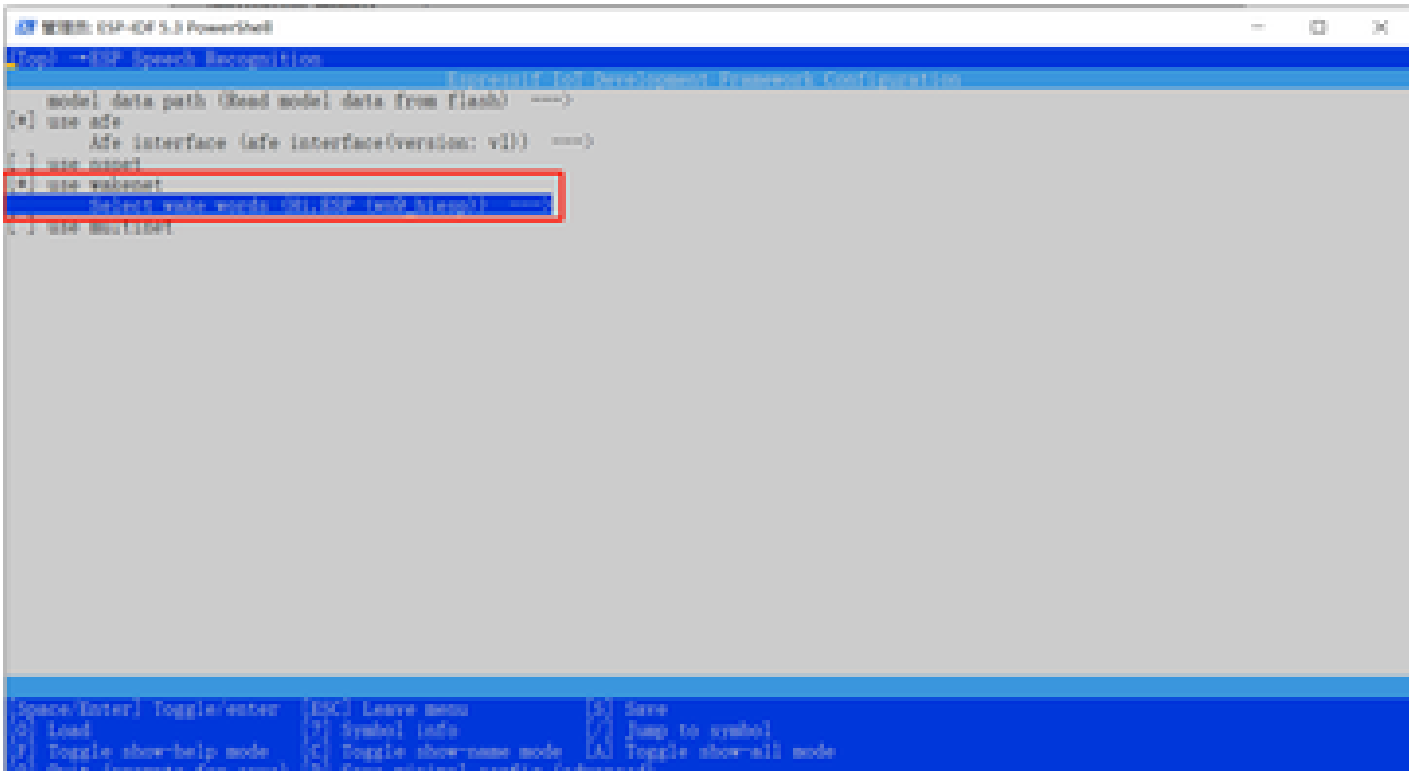
3.2 ESP Speech Recognition Configuration Items

Enter the ESP Speech Recognition configuration item to make configuration changes, as shown in the following figure:

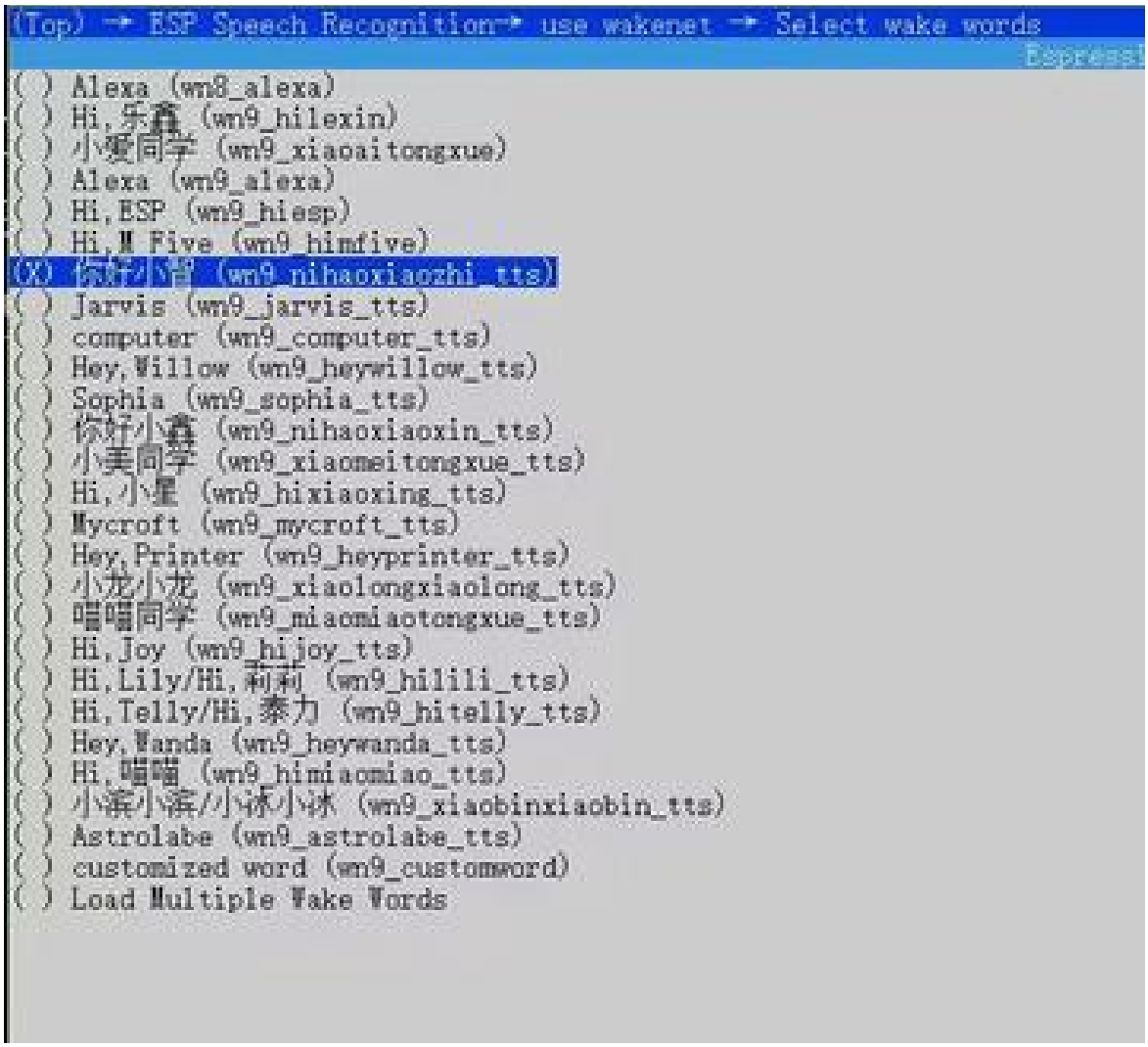


3.3 Enter the wake word configuration in use wakenet

After pressing the Enter key on the keyboard to confirm, enter the configuration page, select Select wake words (xxxxxxxx) under the use wakenet option ---> enter to select and change the wake word.



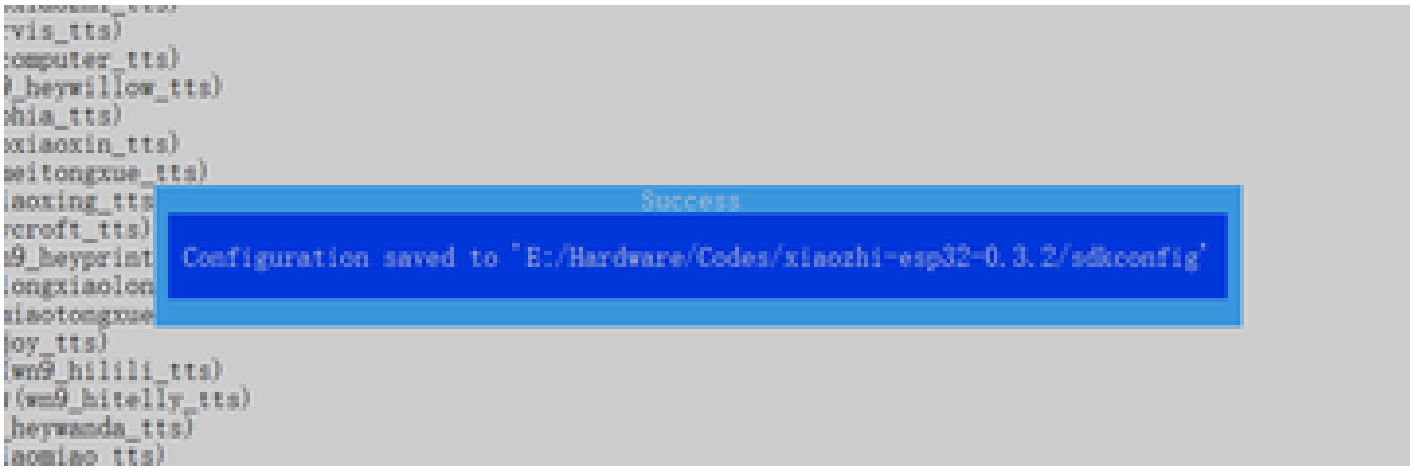
After entering, please select the free wake word you want to use, press Enter to confirm the selection, and an X (X) will appear in the parentheses before the option, indicating selection, as shown in the following figure:



> Note: The above are wake word selection options for the ESP32-S3 as the target chip. If you are using ESP32 or other chips, there will only be a few wake words. Please first use the command idf.py set-target esp32s3 to switch the target chip to esp32s3.

3.4 Save configuration options and exit

After modifying the wake word configuration, you need to save and exit. First, press the keyboard key 【S】 to display the following interface as shown in the figure, then press Enter to Save, the blue prompt layer will disappear, and finally press the keyboard key 【Q】 or Esc to exit the configuration interface.



3.5 Rebuild compilation and flashing

****Note:**** After changing the wake word configuration, you need to recompile the firmware, download it to the ESP32S3 development board, and restart it for the changes to take effect.

After completing the above operations in this section, ensure that the computer is properly connected to the ESP32S3 development board's serial UART interface (the right interface when the TypeC interface on the front of the development board faces downward), and then compile as shown in the following illustration:

- **Rebuild**

Under the root directory of the project source code (taking version xiaozhi-esp32-0.3.2 as an example), xiaozhi-esp32-0.3.2

First, ensure that the target device is esp32s3, and enter the command: idf.py set-target esp32s3 (if it already is, there is no need to perform this step)

Recompile and build the project, enter the command: idf.py build

***Note:** If you have previously compiled this project, to ensure updates, you may choose to use idf.py fullclean to clear the cached components before proceeding with idf.py build

The waiting time for ESP32 program build varies depending on the configuration of the computer's hardware and software environment, generally ranging from a few minutes to 20 minutes. After the project is successfully built, it will prompt "Successfully created esp32s3 image." (If you encounter any issues, please consult in the QQ group), as shown in the following figure.

- **Re-flash and Monitor**

Please ensure that the ESP32S3 development board has been recognized by the computer and confirm the COM Port Number. If there are multiple COM devices on Windows, you can specify a port for flashing and monitoring by entering the command: idf.py -p COM7 flash monitor (if other devices do not support IDF flashing, it's okay not to specify, enter the command: idf.py flash monitor), as shown in the following figure:

Finally, wait for the programming to complete (prompting "Write 100%"), and automatically enter the serial port monitoring mode after completion.

****Note:****

After reflashing the chip firmware of the EPS32S3 development board, if the WIFI version requires reconfiguration of the network, please refer to the previous documentation to configure the network, and then you can wake up using the new wake word.

After changing **the wake word** , when the wake word is input via voice, the serial port monitoring application still returns "Xiaozhi". This does not affect usage and will not be processed for the time being.

**For custom wake words, please consult Espressif official

Espressif officially provides commercial replacement and new word customization services, with relatively high application requirements. For a simple reference, please refer to the following illustration. For details, please visit the Espressif official website.